FLAIR

Federated **L**earning **A**ssessment for **I**CU **R**esearch

A Federated ICU Benchmark for External Validation on the Common Longitudinal ICU Format (CLIF)

CLIF Consortium · FLAIR Working Group

July 6, 2026

MIMIC-CLIF · Rush · Northwestern · UChicago Medicine ▪ 278,065 ICU stays

The question in one line

A model that looks strong on public ICU data —
does it still work at a hospital that **did not build it**?

FLAIR is the federated infrastructure that answers this, reproducibly,
without moving a single patient record.

The gap

Evidence for ICU AI does not keep pace with models

- Models for sepsis, extubation, mortality appear **continually**
- Whether one works *outside* its development hospital is often **unknown**
- The evidence gap is an **access gap**: representative multi-site ICU data rarely reach the team that must test the model

A cautionary anchor

The Epic Sepsis Model — deployed across many US hospitals — fell to **AUC 0.63** on independent external evaluation, against vendor-claimed 0.76–0.83.

Wong et al., JAMA Intern Med 2021; Habib et al. 2021

Sepsis Watch and TREWS each took *years* to reach a handful of sites — serious validation does not scale.

Public benchmarks help — but cannot close the gap alone

- MIMIC, eICU, HiRID, YAIB, EHRSHOT are essential for **reproducible development**
- But they **centralize their test data** — and so:
  - inherit one source's selection biases
  - can *saturate*: tuning against the same public set makes transport look easier than it is
- None offers a standing, ICU-scoped, **federated validation** layer over data that never leaves the hospital

Develop on public MIMIC-CLIF → validate federated.

van de Water et al. 2023 (YAIB); Wornow et al. 2023 (EHRSHOT); Arnrich et al. 2025 (MEDS)

What current ICU benchmarks measure — and miss

- Tasks cluster on **mortality, length-of-stay, and physiologic deterioration** (AKI, sepsis) — even multi-center YAIB stays within mortality / AKI / sepsis / kidney / LOS
- Sparse on **deployment decisions**: disposition (LTACH), extubation failure, readmission, and **leak-safe before-treatment** sepsis (CDC ASE)
- **Global / random splits, not temporal** — no pseudo-prospective, deployment-time estimate
- Centralized (or single-site, e.g. EHRSHOT = one Stanford site) test data — external multi-site validation is not built in, and benchmarks saturate
- Row-level metrics, not *operational* (one alarm per stay)

They measure discrimination on public data — not whether a model *survives deployment* at a new hospital.

van de Water et al. 2023; Wornow et al. 2023; Harutyunyan et al. 2019

Related model-to-data efforts — pieces of the answer

- **OHDSI / LEGEND** — federated network studies over OMOP (share SQL / coefficients)
- **OpenSAFELY** — analytic code to national primary-care data; no patient rows move
- **MedPerf** — model-to-data benchmarking, largely imaging
- **FeTS** — federated *training* for imaging
- **DREAM / EHR DREAM** — model-to-data evaluation as a *challenge*

Missing for ICU EHR: a **standing, ICU-scoped federated benchmark** on a critical-care standard.

FLAIR combines a critical-care standard (CLIF) + a reusable task catalog + a standing federated protocol.

Guinney et al. 2018 (DREAM); Bergquist et al. 2024 (EHR DREAM); OpenSAFELY 2021; Karargyris et al. 2023 (MedPerf); Pati et al. 2022 (FeTS); Reps et al. 2020 (OHDSI/LEGEND)

FLAIR = three coupled artifacts

Data layer	Task layer	Protocol layer
CLIF critical-care standard; public **MIMIC-CLIF** entry point (∼72,800 stays)	Five inspectable Python tasks; **open, PR-extensible** catalog	Committee-reviewed, **aggregate-only** model-to-data execution

The model goes to the data. Only aggregate, cell-suppressed reports come back.

Not a clinical-utility claim — an *evidence layer* between local development and any clinician-facing decision.

Public benchmarks vs FLAIR

Dimension	Public ICU benchmarks	FLAIR
Test data	centralized / single-site	**federated**, stays at each hospital
Tasks	mortality, LOS, AKI, sepsis (generic)	5 **deployment** tasks: LTACH disposition, extubation failure, before-treatment CDC-ASE sepsis, readmission, mortality
Train/test split	global / random	**temporal** (pseudo-prospective) at external sites
External validation	same repository / pooled	**independently governed** hospitals
Scoring	row-level	episodic / landmark / peak (**operational**)
Task catalog	fixed	**PR-extensible**

Complementary, not competing: **develop** on public benchmarks, **validate** federated on FLAIR.

van de Water et al. 2023 (YAIB); Wornow et al. 2023 (EHRSHOT); Pollard et al. 2018 (eICU); Harutyunyan et al. 2019

Methods

Data layer: CLIF and MIMIC-CLIF

- **CLIF** harmonizes ICU EHR content into shared relational tables with mCIDE category definitions (vitals, labs, meds, respiratory support, microbiology, ADT, . . .)
- Task code refers to **shared table names and category values** — portable across sites, minimal per-site feature engineering
- **MIMIC-CLIF** expresses MIMIC-IV in CLIF v2.1 (∼72,800 ICU stays) — the public development set

Develop on public MIMIC-CLIF → validate on private CLIF data.

Rojas et al. 2025 (CLIF); Johnson et al. 2023 (MIMIC-IV)

Three ways to get data out of CLIF

1. Direct feature extraction

Compute features straight from the CLIF tables — the reference baseline's path (count features over an ELF vocabulary).

2. MEDS format *(optional)*

Export as a standardized Medical Event Data Standard event stream for MEDS-native pipelines.

3. Wide dataset

A YAIB-style wide tabular matrix for models that expect one row of features per prediction.

MEDS is **one option**, not a required step — a model developer picks whichever representation fits.

Arnrich et al. 2025 (MEDS); van de Water et al. 2023 (YAIB); Wornow et al. 2023 (EHRSHOT/FEMR)

Task layer: five ICU prediction tasks

Task	Outcome	Grid	Mode	Prev.
T1	ICU daily mortality	daily 07:00	landmark	0.125
T2	LTACH discharge	daily 07:00	landmark	0.062
T3	Extubation failure 24 h	at extubation	episodic	0.054
T4	Sepsis (CDC ASE) 6 h	hourly	peak	0.061
T5	Unplanned ICU readmission	at ICU discharge	episodic	0.093

- One Python module per task: `META` + `build()` $\rightarrow$ one row per prediction opportunity
- Spans outcome, disposition, respiratory, infectious-deterioration, utilization
- Catalog **grows by contribution**: acceptance needs MIMIC reproducibility, clinician face validity, $\geq$2-site sign-off

Task 4 is sepsis by the *surveillance* definition

- **CDC Adult Sepsis Event (ASE)** within 6 h, hourly, hospital-wide
- t_0 = earliest ASE criterion (blood culture, qualifying antibiotics, vasopressor, organ dysfunction) via `clifpy.utils.ase`
- Prediction window ends **strictly before** t_0 — no label leakage from treatment
- Follows the *before-treatment* formulation; contrasts with antibiotic-anchored Sepsis-3

Kamran et al., NEJM AI 2024; Rhee et al. 2019 (CDC ASE)

Why it matters

A rare, hospital-wide, hourly screen — the most clinically consequential task, and the most fragile under distribution shift.

Three safeguards make cohorts comparable across sites

1. **Stitched hospitalizations.** Fragmented episodes (ED→inpatient, discharge→rapid readmit) joined into one `hospitalization_join_id` (6 h gap) — one episode is the unit, never split across train/test
2. **Strict point-in-time.** Features use only data at or before `prediction_dttm` — the no-future-data boundary
3. **Audit hash.** A SHA-256 over task source + policy metadata is stamped into every report

Sites can prove they ran byte-identical task code.

Data splits: pseudo-prospective by design

- **75/25** split at the **stitched-hospitalization** grain — one hospitalization never straddles the boundary
- **Default = temporal split**: oldest 75% train, newest 25% test
- Estimates **pseudo-prospective performance** — how the frozen model behaves *going forward in time*, without a prospective trial
- **MIMIC-CLIF is the exception**: de-identified admission dates carry no usable time order → deterministic hash split

Consequence

At external sites, degradation blends *temporal* shift (train-past → test-future) with *cross-site* shift — both are what a new hospital actually faces.

Reference baseline: one frozen, shippable model

- **XGBoost** over point-in-time **count features** (EHRSHOT/FEMR tradition)
- CLIF events → fixed **211-code** ELF vocabulary; counts of codes strictly before prediction time
- Units never features; discharge diagnoses excluded (no leakage)
- HPO once on MIMIC-CLIF (**Optuna**, 30 trials, 5-fold CV AUROC)
- Model + dictionary **shipped unchanged** to sites for *inference only* — no per-site training

A benchmark check, not a claim that the baseline is clinically adequate. It exists to expose transport behavior.

Wornow et al. 2023; Steinberg et al. 2021 (FEMR)

Three evaluation modes — and what each report contains

Mode	Prediction unit	Report bundle
Episodic (T3, T5)	one prediction per stay	AUROC, AUPRC, ROC & PR curves, **calibration** (slope / intercept / Brier + reliability), **decision-curve analysis**, subgroup AUROC
Landmark (T1, T2)	one per daily landmark	per-landmark $\widehat{\mathrm{AUC}}(t)$ *curve* **plus** a single **pooled dynamic AUC**; lead-time / skill plots
Peak (T4)	stay collapsed to max risk	discrimination + subgroup only; **calibration & DCA omitted** (the "peak-calibration trap")

Every bundle also carries a cohort summary, a suppressed-code registry, and audit metadata. All metrics on the held-out test split; percentile-bootstrap 95% CIs at the *stay* level.

The landmark metric: pooled dynamic AUC

For a stitched hospitalization h with event onset T_h, discharge D_h, horizon τ, at landmark time t:

Risk set (still admitted, event-free):

$$R(t) = \{\, h : T_h > t,\ D_h > t \,\}$$

Dynamic label (event within τ):

$$Y_h(t) = \mathbf{1}\{\, t < T_h \le t + \tau \,\}$$

Predicted risk from history $\mathcal{H}_h(t)$:

$$p_h(t) = \hat{P}(t < T_h \le t + \tau \mid \mathcal{H}_h(t))$$

Tie kernel: $\psi(a, b) = \mathbf{1}\{a > b\} + \frac{1}{2}\mathbf{1}\{a = b\}$

Time-specific AUC:

$$\widehat{\mathrm{AUC}}(t) = \frac{\sum_{i,j \in R(t)} Y_i(t)\,[1 - Y_j(t)]\,\psi(p_i(t), p_j(t))}{\left(\sum_i Y_i(t)\right)\left(\sum_j [1 - Y_j(t)]\right)}$$

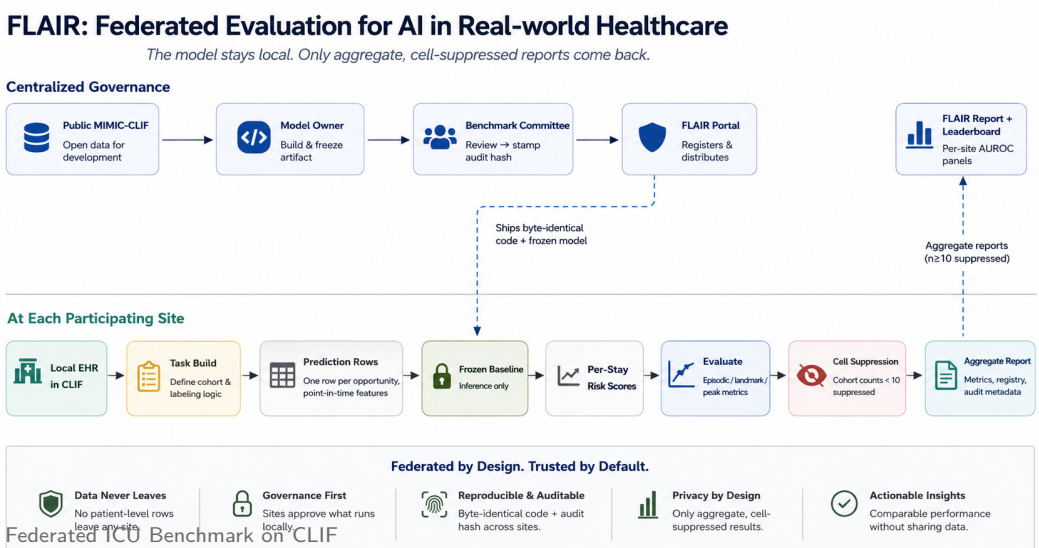
Pooled across landmarks (**reported**):

$$\mathrm{AUC}_{\mathrm{pooled}} = \frac{\sum_t \widehat{\mathrm{AUC}}(t)\, w_t}{\sum_t w_t}, \quad w_t = \mathrm{cases}_t \cdot \mathrm{controls}_t$$

h hospitalization ▪ t landmark ▪ τ horizon ▪ T_h onset ▪ D_h discharge
$R(t)$ risk set ▪ $Y_h(t)$ label ▪ $p_h(t)$ risk ▪ ψ tie kernel ▪ w_t case–control pairs at t

Heagerty & Zheng 2005; van Houwelingen 2007; Blanche et al. 2013

Protocol layer: the model-to-data loop



Privacy and governance are operational, not aspirational

- **No patient-level data leave any site**
- **Cell suppression** enforced by the library: cohort counts < 10 suppressed; subgroups < 10 omitted; sparse calibration bins withheld
- **Governance precedes execution** — committee review; sites keep final approval over what runs locally
- Every report carries: discrimination, calibration/DCA (where applicable), subgroup + lead-time summaries, suppressed code registry, and **audit metadata**

Model-to-data lineage of DREAM / EHR DREAM — FLAIR makes it a *standing ICU benchmark*.

Results

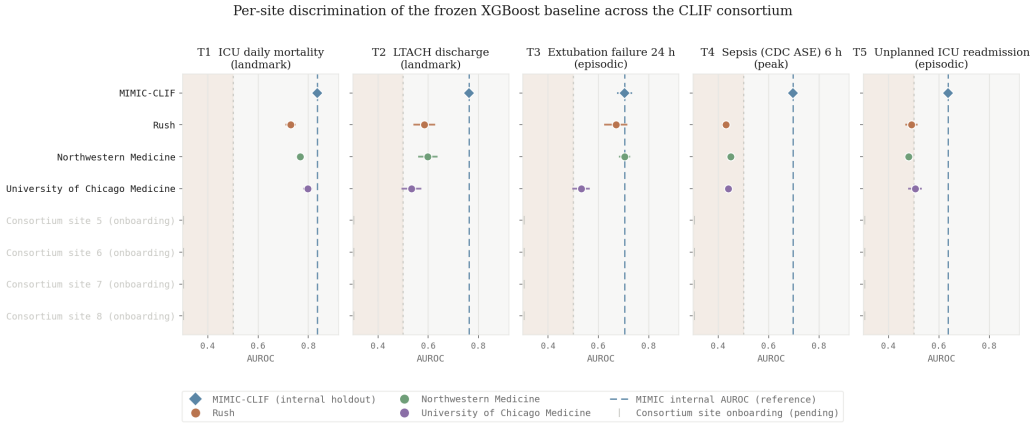
Four sites reporting; four onboarding

Site	ICU stays	F%	Age	SOFA
MIMIC-CLIF	72,778	43.2	67	5
Rush	53,356	47.8	63	4
Northwestern Medicine	114,335	44.0	66	5
UChicago Medicine	37,596	43.5	61	6
Consortium sites 5–8	—	—	—	—

278,065 ICU stays across four independent health systems.

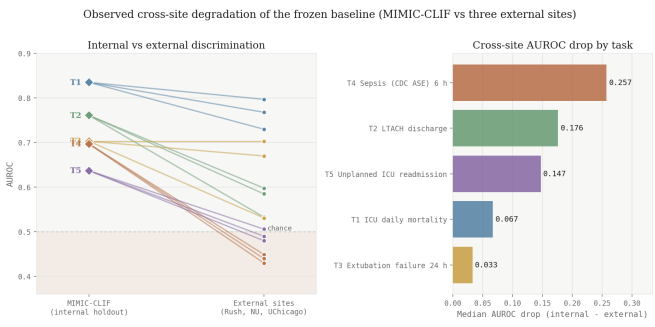
Placeholder rows leave space for the full eight-site consortium — no fabricated numbers.

The headline: per-site discrimination, one panel per task



A single pooled AUROC would hide this. The per-site *spread* is the point.

Strong internal discrimination does not survive transport



Median internal→external AUROC drop:

- Sepsis **0.257**
- LTACH 0.176
- Readmission 0.147
- Mortality 0.067
- Extubation 0.033

Task-dependent, not uniform.

Sepsis falls *below chance* at every external site

MIMIC-CLIF (internal)	**0.697**
Rush	**0.430**
Northwestern Medicine	**0.449**
University of Chicago Medicine	**0.440**

The reproducible external result on the highest-stakes task is a model that is **worse than a coin flip**.

Not a one-hospital artifact — three independent health systems, same failure.

Transportability is task-dependent — and heterogeneous

Transports	Collapses	Splits
Mortality (T1) external 0.73–0.80 vs 0.835 internal	**Sepsis (T4)** below chance everywhere **Readmission (T5)** $\approx$ chance	**Extubation (T3)** Northwestern **0.703** (ties MIMIC) UChicago **0.531**

The within-task, across-site spread is exactly what federated external validation exists to surface.

Discussion

Why federate evaluation — and why now

- **Saturation resistance.** Test data stay siloed — the leaderboard's data never enter public hands, limiting pretraining-on-the-test-set and tune-to-test incentives
- **Evaluation, not training.** Aggregate cell-suppressed metrics leak less than the model updates exchanged in federated *learning*; no per-site model object to share
- **Re-inspectable.** Byte-identical code + audit hashes make cohort definitions, exclusions, and labels auditable across sites

An evidence layer between local development and any clinician-facing claim.

Limitations

- Depends on **CLIF adoption** — unmapped sites need an ETL step first
- Structured EHR + binary classification only — no notes, waveforms, images, or treatment effects
- **Aggregate-only** reporting limits fine-grained error analysis; suppressed small subgroups invisible centrally
- Evaluates **frozen** models — no federated training
- Four sites today (four onboarding); each external site uses a *temporal* split, so some degradation may reflect temporal as well as cross-site shift

Takeaways

1. FLAIR federates ICU **evaluation** on a shared critical-care standard (CLIF) — model-to-data, aggregate-only, auditable
2. A frozen public-data baseline already exposes **large, task-dependent** cross-site degradation across **four** real health systems
3. **Sepsis** discrimination falls *below chance* at every external site; **mortality** transports; **extubation** splits by site
4. The contribution is not a model — it is a **reproducible way to ask** whether ICU models behave alike across institutions

Develop on public MIMIC-CLIF. Validate federated.

Thank you

FLAIR — Federated Learning Assessment for ICU Research

github.com/clif-consortium/FLAIR

MIMIC-CLIF · Rush · Northwestern · UChicago Medicine ▪ 278,065 ICU stays